

Lecture 2 Tue Aug 25 Mini Sverrisson

Present: ~~Self~~ in class/classroom.

Models of Computation

Scope: → Prep

~~Self~~ read. & watched

Marwedel §2.1-2.2 (pp. 29-41); Lee & Seshia ch. 3 (pp. 42-77) → covers Marwedel §2.4 Decks 2.01-2.03, CP-5F videos 03.1-03.3.

Notes:

We talked about thermostats fluctuation with only one temperature threshold. We were asked to build a variant with only one threshold. Instead of hysteresis. The chattering is avoided by time. If the initial state if the temperature $\leq 20^{\circ}\text{C}$ it turns on and leaves it running for 30-seconds. After if the temp is $> 20^{\circ}\text{C}$ it turns the heater off for 2 min. It turns on only if the temp is $\leq 20^{\circ}\text{C}$.

In class:

Marcel the head teacher, ~~Self~~ walked us throug How this would work and play out.

Marwedel §2.1-2.2

- * no modeling language meets all the requirements, everyone is a compromise.
 - * A model of computation = components + how they communicate.
 - * ~~Dependence~~ * Dependence graph: nodes are tasks, edges mean "must finish first"
- Lee & Seshia ch 3

State = a summary of the past: everything that affects how you react to future input.

all in / ~~Witness~~ | Witness: *Kevin* 25.8.2026

Lab 2 - Sensing The Colors.

= Introduction 1.

- Goal: Make a colour classification station, TCS 34725 reads a color, Pi classifies it. SSD1306 Shows the ans.

- ~~Theory~~: I²C → Inter-Integrated Circuit (I²C) BUS.

= Definition: a two-wire, low-speed bus that lets several chips share one pair of open-collector lines, held high by pull-up resistors. (Marwedel, 2021, p. 178).

- Plan: I will SSH into my Pi called FridayPi and follow the steps on:

[@c7641e](https://github.com/mkyas/cpsy-display-ip)

Then I will solder the pins onto TCS 34725 RGB sensor. I do not need to do that to the SSD1306 screen since they come soldered on.

- Design: Figure 1 shows both boards wired to the Pi. Both share SDA (Phys 3 / BCM 2) and SCL (Phys 5 / BCM 3), powered from 3v3 (Phys 1) - GPIO is not 5v tolerant.

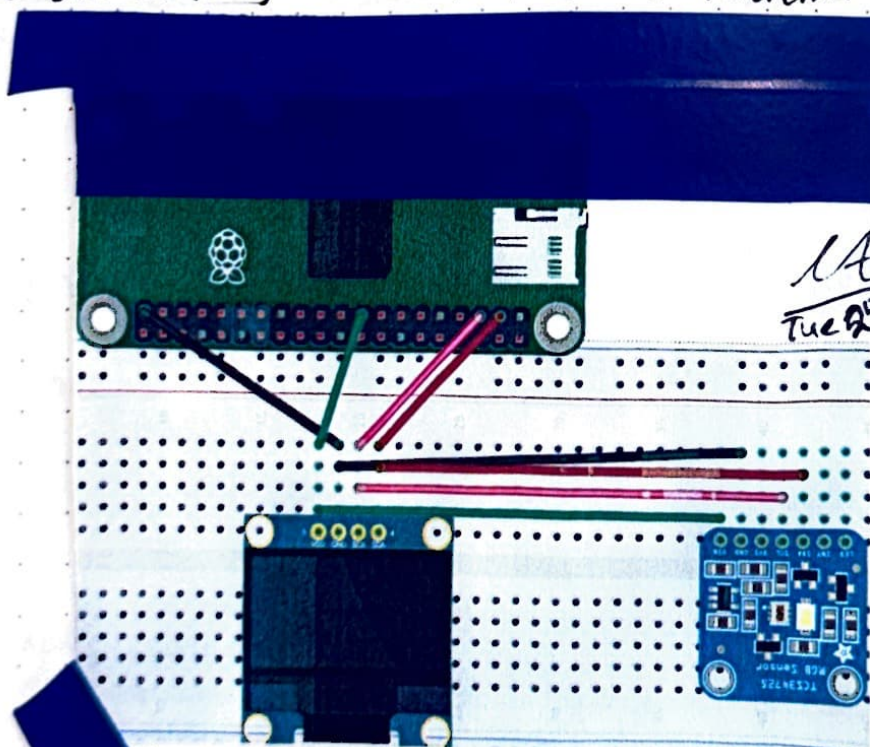


Figure 1: Breadboard wiring, TCS 34725 + SSD1306 Fixing LO.6

Máni

Witness: *Stef*

25.08.26

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lab 2 - Sensing the Colors.